

SHUAI ZHOU

Master Student of Robotics

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EDUCATION

CARNEGIE MELLON UNIVERSITY

Master of Science in Robotics

Advisor: Zackory Erickson

Pittsburgh, USA

Aug 2026 — Jun 2028

SOUTH CHINA UNIVERSITY OF TECHNOLOGY

Bachelor of Engineering in Robotics; GPA: 3.86/4.00

Advisor: Jiaoyang Li

Guangzhou, China

Aug 2022 — Jun 2026

CARNEGIE MELLON UNIVERSITY

Visiting Student in the Robotics Institute

Advisor: Jiaoyang Li, Ding Zhao

Pittsburgh, USA

Jun 2025 — Mar 2026

UNIVERSITY OF CALIFORNIA, BERKELEY

Exchange Student in EECS; GPA: 4.00/4.00

Berkeley, USA

Aug 2023 — Dec 2023

EXPERIENCE

CARNEGIE MELLON UNIVERSITY, RCHI Lab

Research Topics: Dexterous Manipulation, Tool Use.

Advisor: Zackory Erickson

Pittsburgh, USA

Aug 2026 — Present

- Continual learning with failure-aware dexterous manipulation for tool use.

HKUST (Guangzhou), IPRN Lab

Research Topics: Foundation Model, Active Perception, Egocentric Human Data.

Advisor: Haoang Li

Guangzhou, China

Mar 2026 — Jul 2026

- Scaling Active Perception VLA with AMP teleoperation system and Human-Robot co-training [1].
- World action model pretraining with egocentric-human data and corresponding annotation pipeline [2] [5].

CARNEGIE MELLON UNIVERSITY, SAFE AI Lab

Research Topics: Humanoid, Egocentric Human Data, Whole-Body Controller.

Advisor: Ding Zhao

Pittsburgh, USA

Aug 2025 — Mar 2026

- Learning versatile humanoid manipulation policy and developing robust lower-body controller [7].
- Egocentric cross-embodiment manipulation with embodiment-specific latent prediction [4].

CARNEGIE MELLON UNIVERSITY, ARCS Lab

Research Topics: Learning for Planning, Multi-Robot System.

Advisor: Jiaoyang Li

Pittsburgh, USA

Apr 2025 — Mar 2026

- Collaborative multi-robot non-prehensile manipulation via flow-matching model and lazy planning [6].
- Learning execution-time prediction models for closed-loop multi-robot coordination and dependency optimization [3].

SHANGHAI JIAO TONG UNIVERSITY, RAP Lab

Research Topics: Heuristic Search, Multi-Robot System.

Advisor: Zhongqiang Ren

Shanghai, China

Apr 2024 — Apr 2025

- Scalable planners for multi-robot path planning that scale to 1,000+ robots [9].
- Complete, anytime planning framework with pruning and sorting strategies that accelerated search [8].

PUBLICATION (* equal contribution)

PREPRINT

- [1] **ActiveScale: Scaling Active Perception for Robots across Model, Data, and Hardware**
Shuai Zhou*, Kaisheng Pang*, Wenxuan Song, Wenjie Zhang, Xinhua Zheng, Haoang Li. **Under Review, 2026** [Paper] [Project]
- [2] **UniWAM: Unified World-Action Model**
Wenxuan Song, Jiayi Chen, Jingbo Wang, Shuai Zhou, Xicheng Gong, Zehua Fan, Ziyang Zhou, Junwu E, Haodong Yan, Fuhao Li, Qize Yu, Xu Huang, Pengwei Wang, Wen Chen, Shunbo Zhou, Haoang Li **Under Review, 2026** [Paper] [Project]
- [3] **From Discrete Plans to Real-World Execution: A World-Model-Driven Framework for Execution-Aware Multi-Agent Path Finding**
Jingtian Yan, Shuai Zhou, He Jiang, Stephen Smith, Jiaoyang Li. **Under Review, 2026** [Paper]
- [4] **EgoX: Egocentric Cross-Embodiment Manipulation with Embodiment Dreaming**
Yaru Niu*, Binghong Chen*, Changwei Yao*, Shangtao Li*, Zhenlong Fang*, Revanth Krishna Senthilkumaran, Shuai Zhou, Yumein Mao, Kalie Ching, Po-Ju Chao, Yudong Luo, Peiqi Yu, Heng Zhang, Hao Zhang, Bingqing Chen, Chen Qiu, Hongtei (Eric) Tseng, Changliu Liu, Jonathan Francis, Ding Zhao. **Under Review, 2026** [Paper] [Project]

- [5] **EgoANT: Segmentation and Annotation for Egocentric Human Video with open-source VLM**
Shuai Zhou **Blog**, 2026 [\[Blog\]](#)

PUBLISHED

- [6] **LAMP: Long-Horizon Adaptive Manipulation Planning for Multi-Robot Collaboration in Cluttered Space**
Shuai Zhou, Yorai Shaoul, Jiaoyang Li. **IROS**, 2026. [\[Paper\]](#) [\[Project\]](#)
- [7] **HTD: Learning Versatile Humanoid Manipulation with Touch Dreaming**
Yaru Niu, Zhenlong Fang, Binghong Chen, [Shuai Zhou](#), Revanth Krishna Senthilkumaran, Hao Zhang, Bingqing Chen, Chen Qiu, Hongtei (Eric) Tseng, Jonathan Francis, Ding Zhao. **IROS**, 2026. [\[Paper\]](#) [\[Project\]](#)
- [8] **LSRP*: Scalable and Anytime Planning for Multi-Agent Path Finding with Asynchronous Actions**
[Shuai Zhou](#), Shizhe Zhao, Zhongqiang Ren. **SoCS**, 2025. [\[Paper\]](#) [\[Project\]](#)
- [9] **Loosely Synchronized Rule-Based Planning for Multi-Agent Path Finding with Asynchronous Actions**
[Shuai Zhou](#), Shizhe Zhao, Zhongqiang Ren. **AAAI**, 2025. [\[Paper\]](#) [\[Project\]](#)

SKILLS

Platforms: Wuji Hand 2, AgileX ALOHA, AgileX Piper Arm, Apple Vision Pro, Meta Quest 2, Unitree G1 Humanoid

Simulators: MJLab, IsaacLab, MuJoCo, PyBullet

Programming: Python, C++, Java, HTML, MATLAB, $\text{\LaTeX}$

SERVICES

Conference Reviewer: 2025-2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2027 IEEE International Conference on Robotics & Automation (ICRA)

Journal Reviewer: 2026 IEEE Robotics and Automation Letters (RA-L)

HONORS

Merit Student of South China University of Technology

Top student in Robotics Engineering, Class of 2022

Guangzhou, China

Feb 2024

Best Paper Award

IROS 2026 PWMS(Physical World Models for Scaling Embodied AI) Workshop

Pittsburgh, USA

Oct 2026